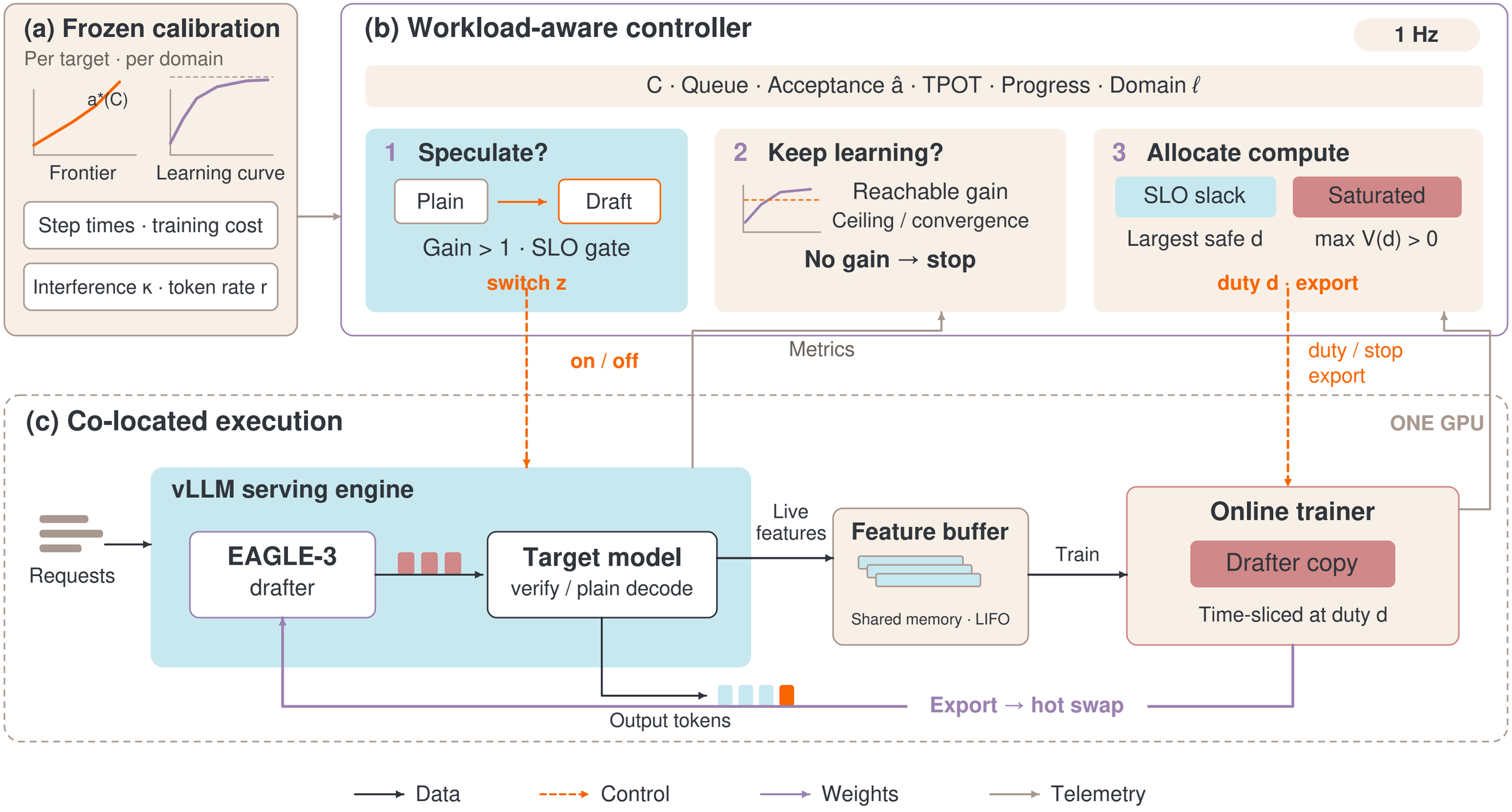
(a) Frozen calibration
Per target · per domain
a*(C)
Frontier
Learning curve
Step times · training cost
Interference κ · token rate r
(b) Workload-aware controller
1 Hz
C · Queue · Acceptance â · TPOT · Progress · Domain ℓ
1 Speculate?
Plain
Draft
Gain > 1 · SLO gate
switch z
2 Keep learning?
Reachable gain
Ceiling / convergence
No gain → stop
3 Allocate compute
SLO slack
Saturated
Largest safe d
max V(d) > 0
duty d
export
on / off
Metrics
duty / stop
export
(c) Co-located execution
ONE GPU
vLLM serving engine
Requests
EAGLE-3
drafter
Target model
verify / plain decode
Live
features
Feature buffer
Shared memory · LIFO
Train
Online trainer
Drafter copy
Time-sliced at duty d
Output tokens
Export → hot swap
Data
Control
Weights
Telemetry